

PHL 1010 Tutorial II

27/08/2025 (Wednesday)

1. (a) A vector field in CCS is expressed as $\vec{V} = y\hat{x} + (x + z)\hat{y}$. If this vector exists at a Cartesian point $(-2, 6, 3)$, what are its components expressed in cylindrical system at the same point?
(b) Consider a vector field given by $\vec{V} = r \sin \phi \hat{r} - (1/r) \sin \theta \cos \phi \hat{\theta} + r^2 \hat{\phi}$. Find out the component of $\vec{V}$ tangential to the surface $r = 10$ at P $(10, 150^\circ, 330^\circ)$.
Further, find out a unit vector at P perpendicular to $\vec{V}$ and tangential to the surface $\theta = 150^\circ$.
 2. Fluid flow is characterized with the following velocity fields:
(i) $\vec{V} = ax\hat{x}$ (ii) $\vec{V} = ay\hat{x}$ (iii) $\vec{V} = \hat{r}/r^2$ (iv) $\vec{V} = \hat{\phi}/r$
(a) Sketch the vector fields.
(b) Calculate divergence and curl (use appropriate coordinate system).
 3. Problems from Griffiths (3rd edition):
1.33, 1.42, 1.43, 1.53, 1.55, 1.58
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